Synthesis of Heptyl Acetate via Fischer Esterification

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1 Introduction

The objective of this experiment is to synthesize heptyl acetate via Fischer Esterification by reacting 1-heptanol with acetic acid in the presence of catalytic concentrated sulfuric acid. The product will be characterized and analyzed by odor analysis, ATR-IR, 100 MHz ^{1}H NMR, and GC-MS.

2 Reaction Scheme

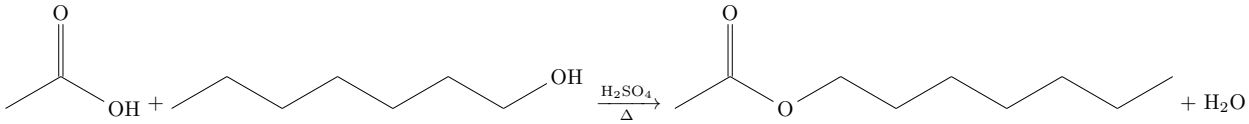


Figure 1: Reaction of Acetic Acid and 1-Heptanol

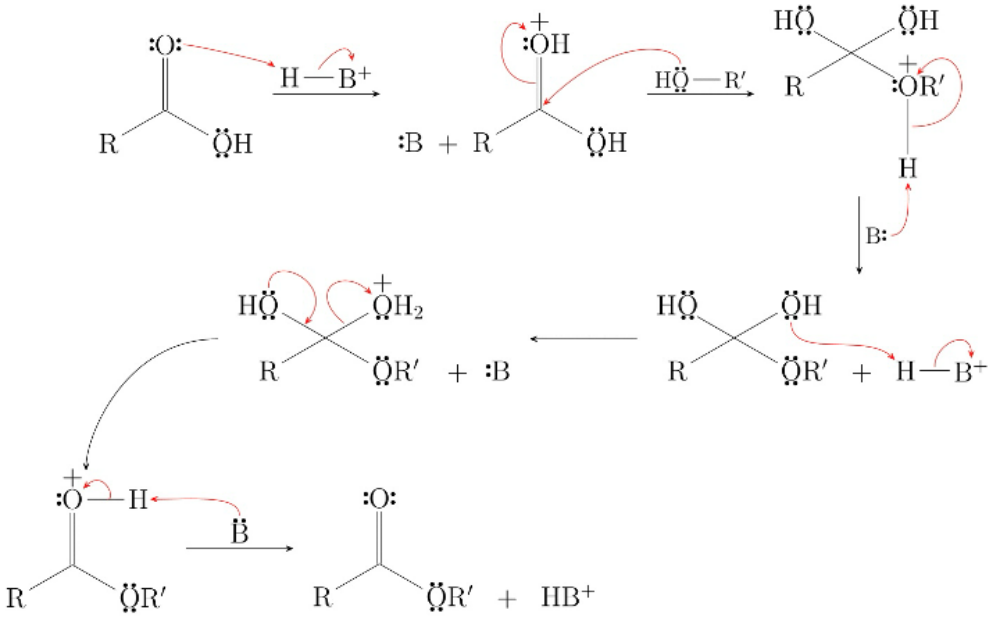


Figure 2: Fischer Esterification Mechanism

3 Table of Reagents and Products

Compound	MW (g/mol)	Density (g/mL)	B.P (°C)	Equiv	Amount	mmol
Acetic Acid	60.05	1.049	118.1	1.2	2.78 g / 2.65 mL	46.3
1-Heptanol	116.20	0.822	176-177	1.0	4.482 g / 5.45 mL	38.6
Sulfuric Acid	98.08	1.84	–	cat.	3 - 4 drops	–
Cyclohexane	84.16	0.779	80.7	–	70 g / 90 mL	–
Heptyl Acetate	158.24	0.872	191-193	–	6.1 g / 7.0 mL	–

4 Experimental Procedure

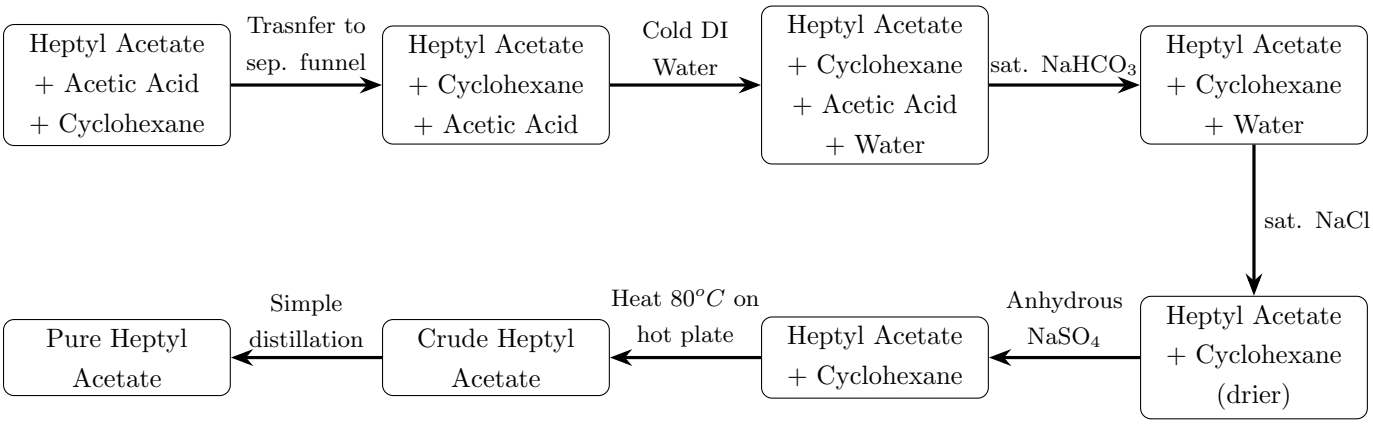
4.1 Apparatus

- Dean Stark
- Simple Distillation

4.2 Procedure

- In a 50 mL round-bottom flask, add 6.403 g (7.79 mL) of 1-heptanol followed by 6.618 g (6.31 mL) of acetic acid and a boiling stone.
- While stirring, slowly add 3-4 drops of concentrated sulfuric acid.
- Gently reflux for 60 minutes, then allow the mixture to cool to room temperature.
- Chill the product in an ice bath then transfer the mixture to a separatory funnel using 20 mL total of diethyl ether to aid in the process.
- Quench the organic layer with 15 mL of iced cold DI water and collect the organic layer. Do this twice.
- Neutralize and extract the organic layer with small portions of a saturated sodium bicarbonate solution until the production of CO_2 gas ceases.
- Wash the remaining organic layer with a brine solution then dry the organic over anhydrous sodium sulfate followed by gravity filtration.
- Gently boil off the ether solution in a 50 mL beaker blowing a light stream of air over the top to obtain the crude hexyl acetate.
- Purify the product with regular distillation.
- Analyze the purified product by odor assessment, ATR-IR, ^{1}H NMR (100 MHz), and GC-MS to confirm ester formation and purity.

5 Purification Scheme



6 Results and Data

g of Acetic Acid	g of 1-Heptanol	Drops of H_2SO_4	Reflux Time
Mass of Crude Product		Mass of Purified Product	
Theoretical Yield			

Crude Percent Yield	Pure Percent Yield
Crude Product / Procedural Observations	Purified Product Observations